SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*
Assessment Tool Weightage: 40%

Annexure A
SHORT ANSWER TEST

Description about the assessment tool:
Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Code of Conduct:

I GLEONA ALIS. J (192511088) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Leona

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	2	2	2	2
Organization	15%	1½	1½	1	1
Accuracy	10%	1	1	1	1
Clarity	5%	½	½	½	½

Total Marks 34 **/40**

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and	Mostly accurate with minor	Partially correct with	Incorrect answers with

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted,

		covers all required points	omissions	missing points	major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

2) TRANSPORT LAYER - SLIDING WINDOW PROTOCOL

Given:

- Sliding window size = 6 frames
- Each frame = 1024 bytes
- Total frames = 48

Number of transmission windows.

$$\text{Transmission windows} = \frac{48}{6}$$
$$= 8$$

Total transmissions:

Since 1 frame is lost in every window, one ~~transmission~~ is required for each window.

Retransmission = 8

→ Transmission windows required = 8
→ Total transmission = 8

Theory:

- Flow control is a transport layer mechanism that controls the rate of data transmission between sender and receiver

• It ensures that the sender doesn't send data faster than a receiver can process it.

• This prevents receiver buffer overflow, reduces packet loss and retransmissions, improves bandwidth utilization.

3) NETWORK LAYER - IP FRAGMENTATION

Given:

- IP packet size = 12,000 bytes
- MTU = 1500 bytes
- IP header = 20 bytes

Calculation:

1500 - 20 = 1460 bytes

Number of fragments

$$\frac{1200}{1480} = 8.11$$

$$\simeq 9$$

Header overhead

9 x 20 = 160 bytes

Answer:

TOTAL FRAGMENTS = 9

HEADER OVERHEAD = 160 bytes

Theory:

→ When an IP packet is larger than the MTU, the network layer divides it into smaller fragments.

→ Each fragments contains its own IP header with field such as identification, fragment offset, and more fragment flag.

→ These field allow the destination device to correctly identify, arrange and resemble all fragments into the original packet ensuring successfull packets delivery.

4) TRANSPORT LAYER - SLIDING WINDOW PROTOCOL

Transmission windows

$$\frac{48}{6} = 8$$

One frame lost in every window

$1 \times 8 = 8$

Answer:

Transmissions window = 8

Retransmissions = 8

Theory:

→ Flow control regulates the amount of data transmitted by the sender according to the receiver's capacity

→ It prevents congestion, avoids data loss due to buffer overflow, minimizes unnecessary retransmissions and improves the speed, reliability.

5) SESSION LAYER - SYNCHRONISATION

Given:

- File size = 600 MB
- Checkpoint after every 60 MB
- Failure occurs after 390 MB

CALCULATION

Number of checkpoints before failure

checkpoints are created at:

60 MB, 120 MB, 180 MB, 240 MB, 300 MB,

360 MB

Total checkpoints = 6

Data to be transmitted

390 - 360 = 30 MB

Answer:

→ checkpoints created = 6

→ Data to be retransmitted = 30 MB

Theory:

• The session layer uses synchronisation points during communication to support recovery from failures.

• If transmission is interrupted, the communication resumes from the last saved checkpoint instead of restarting from the beginning.

- This reduces retransmission time, saves network bandwidth, improves reliability, makes large data transfers more efficient.